

Chrome Enterprise – Policy & Auto-Update Cheat Sheet

Overview

This guide explains how to configure, secure, and persist Chrome Enterprise policies on macOS using plist files. It focuses on disabling Chrome's auto-update mechanism and enforcing enterprise configuration persistence.

1. Policy Files and Their Purpose

Chrome Enterprise uses two key property list (plist) files on macOS:

- **com.google.Chrome.plist**: Controls browser-level settings (e.g., component updates, extension policy, security options). - **com.google.Keystone.plist**: Controls Google Software Update (Keystone), which handles auto-updates for Chrome and other Google apps.

2. Recommended File Locations

```
/Library/Managed Preferences/com.google.Chrome.plist  
/Library/Managed Preferences/com.google.Keystone.plist
```

These are system-level locations. Chrome reads from `Library/Managed Preferences` first for enterprise configurations. If deployed via MDM, macOS automatically manages and re-creates these files when deleted.

3. Example Configurations

Disable Chrome Auto Updates (Keystone)

```
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">  
<plist version="1.0">  
  <dict>  
    <key>updatePolicies</key>  
    <dict>  
      <key>global</key>  
      <dict>  
        <key>UpdateDefault</key>  
        <integer>3</integer>  
      </dict>  
    </dict>  
  </dict>  
</plist>
```

Disable Component Updates and Enforce Extension Manifest V3

```
<?xml version="1.0" encoding="UTF-8"?>  
<!DOCTYPE plist PUBLIC "-//Apple//DTD PLIST 1.0//EN" "http://www.apple.com/DTDs/PropertyList-1.0.dtd">  
<plist version="1.0">  
  <dict>  
    <key>ComponentUpdatesEnabled</key>  
    <false/>  
    <key>ExtensionManifestV2Availability</key>  
    <integer>2</integer>
```

```
</dict>
</plist>
```

4. Ownership and Permissions

```
sudo chown root:wheel /Library/Managed\ Preferences/com.google.Chrome.plist
sudo chown root:wheel /Library/Managed\ Preferences/com.google.Keystone.plist
sudo chmod 644 /Library/Managed\ Preferences/com.google.*
```

Files must be owned by root:wheel for Chrome to accept them as managed policies.

5. Apply and Refresh

```
sudo killall cfprefsd
```

This command clears macOS preference caches so that Chrome reloads the new policies. Afterward, restart Chrome and visit `chrome://policy`` to confirm that the policies appear with status "Platform".

6. Lock Files (Immutable Flag)

```
sudo chflags schg /Library/Managed\ Preferences/com.google.Chrome.plist
sudo chflags schg /Library/Managed\ Preferences/com.google.Keystone.plist
```

Setting the system immutable flag prevents even root or update agents from deleting or modifying these files. To modify them later, clear the flag first:

```
sudo chflags noschg /Library/Managed\ Preferences/com.google.Chrome.plist
```

7. Verification

```
defaults read /Library/Managed\ Preferences/com.google.Keystone
defaults read /Library/Managed\ Preferences/com.google.Chrome
```

Confirm that the keys and values appear correctly. In Chrome, go to `chrome://policy`` → click "Reload policies". You should see `UpdateDefault = 3`` (manual updates only) and `ComponentUpdatesEnabled = false``.

8. Recovery & Maintenance

If the plist files are deleted during Chrome or system updates: 1. The immutable flag prevents deletion. 2. If removed manually, restore from a backup using a LaunchDaemon or configuration profile. 3. Consider wrapping these plists in a signed .mobileconfig for permanent, MDM-compliant enforcement.